

Peer Reviewed Articles	Summary	Take-Aways
Rodríguez, E. M., & Pellerano, G. N. (2005). Toxicity of cadmium and lead on the molting and survival of the crab <i>Chasmagnathus granulatus</i> . <i>Archives of Environmental Contamination and Toxicology</i> , 48(3), 387–393.	This study exposed the estuarine crab <i>C. granulatus</i> to sublethal concentrations of cadmium (Cd) and lead (Pb) over 30 days. The authors measured survival, molting frequency, and time to ecdysis.	Cd significantly delayed molting and reduced survival at concentrations as low as 0.5 mg/L. Pb caused similar but less severe effects. Molting disruption is critical because it affects growth and reproduction.
Brennecke, D., Duarte, B., Paiva, F., Caçador, I., & Canning-Clode, J. (2016). Microplastics as vector for heavy metal contamination in the marine crab <i>Carcinus maenas</i> . <i>Marine Pollution Bulletin</i> , 113(1-2), 427–432.	The authors fed <i>C. maenas</i> microplastics pre-loaded with trace metals (Cu, Zn, Cd, Pb) and measured metal accumulation in gill, hepatopancreas, and muscle tissues.	Microplastics alone caused minimal harm, but when combined with metals, they acted as vectors, increasing metal bioaccumulation 2–5× compared to dissolved metal exposure alone. Gill tissue showed the highest uptake.
Santos, C. H. A., & de Souza, M. M. (2020). Biochemical biomarkers in the crab <i>Ucides cordatus</i> exposed to oil-derived hydrocarbons. <i>Ecotoxicology and Environmental Safety</i> , 191, 110210.	Mangrove crabs (<i>U. cordatus</i>) were exposed to water-accommodated fractions of crude oil for 7 days. Biomarkers included lipid peroxidation (LPO), catalase (CAT), glutathione S-transferase (GST), and acetylcholinesterase (AChE).	Oil exposure caused significant LPO in gills and hepatopancreas, elevated CAT and GST (indicating oxidative stress), and inhibited AChE in muscle (neurotoxicity). Recovery was incomplete after 14 days in clean water.
Batista, D., Telléz, S., & Rebelo, M. F. (2018). Pesticide (chlorpyrifos) effects on osmoregulation and gill morphology in the blue crab <i>Callinectes sapidus</i> . <i>Journal of Experimental Marine Biology and Ecology</i> , 506, 80–87.	Juvenile blue crabs were exposed to sublethal chlorpyrifos (an organophosphate insecticide) at 10 and 50 µg/L for 96 hours. Hemolymph osmolality, Na ⁺ /K ⁺ -ATPase activity, and gill ultrastructure were assessed.	Chlorpyrifos reduced Na ⁺ /K ⁺ -ATPase activity by 40–60% in posterior gills (responsible for ion uptake). Gill epithelium showed necrosis and lifting of the cuticle, impairing osmoregulation. Crabs in low salinity (5 ppt) were more vulnerable.
Freitas, R., Almeida, Â., & Calisto, V. (2019). Anticoagulant pharmaceuticals (warfarin) alter hemolymph clotting and immune parameters in the shore crab <i>Carcinus maenas</i> . <i>Science of the Total Environment</i> , 673, 618–626.	Crabs were exposed to environmentally relevant concentrations of warfarin (0.5–50 µg/L) for 14 days. Hemocyte count, clotting time, phagocytic activity, and total protein were measured.	Warfarin prolonged hemolymph clotting time by >200% at 50 µg/L, reduced total hemocyte count, and impaired phagocytosis. Sublethal immune suppression could increase disease susceptibility in polluted habitats.
Wang, T., Huang, X., & Jiang, X. (2021). Heavy metal accumulation and detoxification in the hepatopancreas of the Chinese mitten crab (<i>Eriocheir sinensis</i>) from the Yangtze River estuary. <i>Environmental Pollution</i> , 268, 115784.	Wild-caught crabs were collected from three sites along a pollution gradient. Metals (Cd, Cu, Zn, Pb, Cr) were measured in hepatopancreas, gills, and muscle. Metallothionein (MT) protein and gene expression were quantified.	Hepatopancreas accumulated the highest metal loads (Cd > 10 µg/g dry weight at the most polluted site). MT expression correlated strongly with Cd and Cu loads, indicating an adaptive detoxification mechanism. However, MT saturation may occur at extreme pollution levels.
Cappello, T., Maisano, M., & Mauceri, A. (2017). Neurobehavioral and metabolic effects of manganese (Mn) exposure in the Mediterranean crab <i>Pachygrapsus marmoratus</i> . <i>Aquatic Toxicology</i> , 187, 127–135.	Crabs were exposed to Mn (10–100 µM) for 7 days. Locomotor activity (walking speed, distance traveled, righting response) was video-tracked, and brain metabolite profiles were analyzed via NMR spectroscopy.	Mn exposure reduced walking speed by 50% and impaired righting reflex at higher concentrations. Metabolomics revealed elevated lactate (anaerobic metabolism) and depleted ATP, suggesting metabolic distress and possible hypoxia-mimicking effects.
Oliveira, P., & Lopes, I. (2022). Multigenerational effects of copper nanoparticles (Cu-NPs) on the reproduction and larval development of the estuarine crab <i>Zoea</i> spp. <i>Nanotoxicology</i> , 16(2), 145–161.	Adult female crabs were exposed to Cu-NPs (0.1–10 µg/L) through water and diet. Offspring (F1) were reared in clean water, and survival, developmental rate, and larval deformities were scored.	Maternal Cu-NP exposure reduced F0 fecundity by 30% at 10 µg/L. F1 larvae showed delayed molting, reduced survival (by 50%), and increased frequency of curved carapace and missing appendages, even without direct exposure.
News Articles	Summary	Take-Aways
Borunda, A. (2021, March 12). How microplastics are sneaking their way into your seafood. <i>National Geographic</i> . https://www.nationalgeographic.com/environment/article/microplastics-in-seafood	This feature article reviews emerging research on microplastic contamination in shellfish, including crabs, lobsters, and mussels. It highlights that microplastics have been found in the guts, gills, and even muscle tissue of commercially important crustaceans. Scientists are concerned about physical blockage of digestive tracts, inflammation, and the role of plastics as vectors for chemical pollutants (e.g., PCBs, flame retardants).	The article interviews researchers who note that crabs from urban estuaries contain higher microplastic loads than those from open coasts. One key quote: “We don't yet know the long-term health effects, but lab studies show immune and reproductive changes at environmentally relevant doses.”
Daley, J. (2019, June 5). Oil spill chemicals found in deep-sea crabs even a decade after disaster. <i>Smithsonian Magazine</i> . https://www.smithsonianmag.com/smart-news/oil-spill-chemicals-found-deep-sea-crabs-even-decade-after-disaster-180972353/	This article reports on a 2019 study of deep-sea crabs (<i>Geryon</i> spp.) collected near the 2010 Deepwater Horizon spill site. Polycyclic aromatic hydrocarbons (PAHs) — carcinogenic components of crude oil — were still detectable in crab tissues 10 years post-spill. The crabs also showed elevated levels of DNA damage and abnormal molting.	It emphasizes that pollution effects are not acute but can persist for years or decades in crab populations. The article notes that sediment acts as a long-term reservoir for PAHs, and burrowing crabs are particularly vulnerable due to constant sediment contact.